Shreyaa Jayant Dani

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Education

University of Massachusetts Amherst — **Amherst, MA**
Master of Science in Computer Science – GPA: 4.0/4.0 — *Expected May 2026*

Vellore Institute of Technology — **Bhopal, India**
Bachelor of Technology in Computer Science and Engineering – GPA: 8.7/10 — *May 2022*

Technical Skills

Languages: Python, Java, C#, SQL, JavaScript, TypeScript
AI/ML: PyTorch, TensorFlow, scikit-learn, XGBoost, OpenCV, spaCy, LLM Integration, RAG Pipelines, OpenAI API, Google Gemini API, Prompt Engineering, Model Evaluation, Entity Extraction, Feature Engineering
Data Engineering: Snowflake, ETL Pipelines, Data Validation, PostgreSQL
Cloud & Infrastructure: AWS (EC2, S3, DynamoDB, Neptune), Docker, CI/CD, Linux
Web Development: FastAPI, Django REST Framework, React, GraphQL, REST APIs
Tools & Methods: Git, OCR Preprocessing, Fuzzy Matching, Agile, Model Deployment

Experience

Cisco — **Amherst, MA**
Graduate Research Extern — *Jan 2026 – Present*

- Research adaptive monitoring frameworks to detect evasive sabotage in autonomous LLM agents under **Prof. Andrew McCallum** in collaboration with Cisco. Developed **ReAct-based reflective monitor** evaluated on SHADE-Arena against Monitor Red-Teaming baselines,addressing AI safety challenge where static monitors fail to detect subtle, long-horizon misbehavior

Chewy Inc. — **Boston, MA**
AI Innovation Intern II — *Jun 2025 – Dec 2025*

- **Vet Prescription Intelligence (Sep – Dec 2025)**
- Built and productionized an OCR + LLM prescription extraction system with reproducible training pipelines processing 14K+ weekly authorizations with 94% accuracy, utilizing advanced image preprocessing (orientation detection, adaptive sharpening, handwritten text enhancement)
- Designed entity extraction pipeline for client, pet, clinic, and veterinarian data with refill logic and created validation scoring engine using fuzzy string matching and Snowflake pharmacy metadata
- Automated duplicate detection and workflow routing, delivering **$1.4M in annual labor savings**; led UAT and pilot launch with veterinary teams achieving 92% user satisfaction
- **Pharmacy Intelligence (Jun – Aug 2025)**
- Standardized and analyzed **3M+ pharmacy records** to power unified analytics and downstream LLM workflows; engineered multi-layered clinical knowledge graph in AWS Neptune enabling prescription clustering, summary generation, and medicine prediction, **reducing vet review time by 10 times**
- Built LLM-driven veterinary assistant chatbot integrating Snowflake and Confluence data for privacy-compliant, real-time clinical recommendations
- Trained and deployed XGBoost model achieving **95.5% accuracy and 0.95 F1-score** for prescription approval prediction; developed full-stack analytics dashboard using Python, TypeScript, GraphQL, and React

IBM India Pvt. Ltd. — **Pune, India**
Associate Developer — *Jan 2022 – Jun 2024*

- Developed C#.NET backend features for enterprise document workflow systems serving **5000+ users** and implementing transactional tracking, schema updates, and dynamic form generation
- Engineered DB2–SQL Server integration pipelines enabling real-time data synchronization across production environments; improved data consistency by 40%
- Built proactive job-failure alerting system with templated notifications, reducing mean time to resolution (MTTR) by 35% and Provided L2/L3 production support for 40+ enterprise applications

Projects

UMunch – AI Wellness Assistant | *React Native, FastAPI, Snowflake, Google Gemini, ElevenLabs* — **HackUMass XIII**

- Developed cross-platform AI wellness app with allergen-aware meal recommendations and voice-based food logging serving 100+ beta users during hackathon demo Link

Knowledge Distillation for Student Ensembles | *PyTorch, Python, NVIDIA RTX 4070 GPU* — **Academic Project**

- Developed hybrid Knowledge Distillation framework reducing model parameters by 33% while achieving superior accuracy on CIFAR-10 image classification tasks
- Trained and evaluated student-teacher architectures using PyTorch, analyzing model compression efficiency and transfer learning performance to improve deployment readiness Repo